

Lecture 6: Aug 29

Last time

- Calculus of Probabilities (1.2)

Today

- no class next Monday (Labor day)
- Redo Theorem 1 proof
- Binomial theorem
- Conditional Probability (1.3)
- Independence (1.3)

Theorem If $\Pr$ is a probability function and A and B are any sets in $\mathcal{B}$, then

1. $\Pr(B \cap A^c) = \Pr(B) - \Pr(A \cap B)$;
2. $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$;
3. If $A \subset B$, then $\Pr(A) \leq \Pr(B)$.

proof:

Formula (2) in the above theorem gives a useful inequality for the probability of an intersection (Bonferroni's Inequality):

$$\Pr(A \cap B) \geq \Pr(A) + \Pr(B) - 1.$$

Theorem If $\Pr$ is a probability function, then

1. $\Pr(A) = \sum_{i=1}^{\infty} \Pr(A \cap C_i)$ for any partition $C_1, C_2, \dots$;
2. $\Pr(\cup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \Pr(A_i)$ for any sets $A_1, A_2, \dots$

where (1) is also referred to as “Total probability” and (2) is Boole's inequality.

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Conditional Probability

All of the probabilities that we have dealt with thus far have been unconditional probabilities. A sample space was defined and all probabilities were calculated with respect to that sample space. In many instances, however, we are in a position to update the sample space based on new information. In such cases we want to be able to update probability calculations or to calculate *conditional probabilities*.

Definition If A and B are events in S , and $\Pr(B) > 0$, then the *conditional probability* of A given B , written $\Pr(A|B)$, is

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}.$$

Note that B becomes the sample space now: $\Pr(B|B) = 1$.

Example Four cards are dealt from the top of a well-shuffled deck. What is the probability that they are the four aces? What is the probability of getting four aces at the top if knowing the first card is an ace? (there are in total 52 cards)

solution:

Theorem (Bayes' Rule) Let $A_1, A_2, \dots$ be a partition of the sample space, and let B be any set. Then, for each $i = 1, 2, \dots$,

$$\Pr(A_i|B) = \frac{\Pr(B|A_i) \Pr(A_i)}{\sum_{j=1}^{\infty} \Pr(B|A_j) \Pr(A_j)}.$$

proof: